American Journal of Epidemiology
Copyright © 2003 by the Johns Hopkins Bloomberg School of Public Health
All rights reserved
Vol. 157, No. 2
Printed in U.S.A.
DOI: 10.1093/aje/kwf181

Case-Control Study of Physical Activity and Breast Cancer Risk among Premenopausal Women in Germany

Karen Steindorf[1], Martina Schmidt[1], Silke Kropp[2], and Jenny Chang-Claude[2]

[1] Unit of Environmental Epidemiology, German Cancer Research Center, Heidelberg, Germany.
[2] Division of Clinical Epidemiology, German Cancer Research Center, Heidelberg, Germany.

Received for publication June 3, 2002; accepted for publication August 5, 2002.

Important aspects of the recognized inverse relation between physical activity and breast cancer risk are still under discussion. Data on physical activity from sports, occupational activity, household tasks, walking, and cycling by reported frequency, duration, and intensity during adolescence and young adulthood were collected in 1999–2000 from 360 premenopausal breast cancer cases and 886 controls who had previously participated in a German population-based case-control study. In multivariate conditional logistic regression, no association between total physical activity and premenopausal breast cancer was found in two age periods. For women who were active during both periods, the adjusted odds ratio was 0.83 (95% confidence interval: 0.60, 1.14). When both age periods were combined, higher quartiles of total physical activity compared with the lowest quartile showed adjusted odds ratios of 0.97, 0.68, and 0.94. Only the effect of moderately high physical activity was statistically significant. Analyses by type of activity revealed significant protective effects for women who reported the highest levels of cycling activities (adjusted odds ratio = 0.66, 95% confidence interval: 0.45, 0.97). These data do not suggest an inverse monotonic association between total physical activity and breast cancer risk in premenopausal women. The study prevalence of cycling and walking for transportation demonstrated that national habits need consideration in the exposure assessment.

breast neoplasms; case-control studies; exercise; premenopause; questionnaires

Abbreviations: CI, confidence interval; MET, metabolic equivalent; OR, odds ratio.

Many studies have examined the role of physical activity in the etiology of breast cancer. Nine (1–9) of 17 cohort studies (1–17) and 15 (18–32) of 26 case-control studies (18–43) that included more than 100 cases demonstrated a significant inverse association among the most physically active participants compared with the least active. A further six studies, one cohort (13) and five case-control (33, 35, 38–40), can be classified as supportive for an inverse association. A comprehensive review of the literature up to the year 2000 is presented by Friedenreich (44). The International Agency for Research on Cancer (45) recently concluded that there is sufficient evidence in humans for a preventive effect of physical activity on breast cancer risk.

Nevertheless, important aspects of the recognized inverse relation are still under discussion (45, 46). One open question is whether physical activity influences breast cancer risk differentially for pre- and postmenopausal women (21, 30, 46, 47). Evidence from studies investigating premenopausal breast cancer is not as consistent as for total breast cancer. Of the 20 studies (5, 16, 18, 20, 22, 25, 26, 29–34, 37–40, 42, 43, 47) that provided information on premenopausal breast cancer risk, only six (5, 18, 20, 26, 31, 38) clearly demonstrated an inverse association among the most physically active participants compared with the least active.

To elaborate on concrete public health recommendations, further clarification of the association between physical activity and breast cancer is needed. We therefore analyzed data from a recently conducted case-control study with the following specific aims: 1) to obtain all-day coverage of physical activity including sports, household tasks, transportation (walking and cycling), and occupational activity during adolescence and early adulthood; and 2) to analyze whether the two age periods or specific activity patterns over

Correspondence to Dr. Karen Steindorf, Unit of Environmental Epidemiology, German Cancer Research Center, Im Neuenheimer Feld 280, D-69120 Heidelberg, Germany (e-mail: k.steindorf@dkfz-heidelberg.de).

TABLE 1. Characteristics and potential breast cancer risk factors for cases and controls and for physically inactive controls in a population-based case-control study (*n* = 1,246) of premenopausal breast cancer in Germany, 1999–2000

Characteristic	Cases		Controls		Controls only: proportion classified as physically inactive*	
	No.	%	No.	%	No.	Row %
Total	360	100	886	100	123	13.9
Study region						
Rhein-Neckar-Odenwald	256	71.1	600	67.7	87	14.5
Freiburg	104	28.9	286	32.3	36	12.6
Age (years) at diagnosis/recruitment						
<35	37	10.3	104	11.7	13	12.5
35–39	64	17.8	158	17.8	21	13.3
40–44	107	29.7	292	33.0	40	13.7
≥45	152	42.2	332	37.5	49	14.8
Educational level						
Low	37	10.3	97	11.0	4	4.1†
Medium	240	66.7	546	61.6	68	12.5
High	83	23.1	243	27.4	51	21.0
Family history of breast cancer						
No	309	85.8	836	94.4†	118	14.1
Yes	51	14.2	50	5.6	5	10.0
Lifetime smoking						
Never smoker	166	46.1	423	47.7	65	15.4
Former smoker	75	20.8	212	23.9	33	15.6
Current smoker	118	32.8	251	28.3	25	10.0
Data missing	1	0.3	0		0	
Oral contraceptive use						
Never	57	15.8	168	19.0	18	10.7
Ever	303	84.2	718	81.0	105	14.6
Body shape at menarche						
Lean	223	61.4	511	57.7	69	13.5
Normal	120	33.3	330	37.3	51	15.5
Full figured	12	3.3	36	4.1	3	8.3
Data missing	5	1.4	9	1.0	0	

Table continues

time reveal differential effects of physical activity on premenopausal breast cancer risk.

MATERIALS AND METHODS

The present study, conducted in 1999–2000, was based on a population-based breast cancer case-control study carried out in 1992–1995 in two regions of southern Germany: the area of Rhein-Neckar-Odenwald and the greater area of Freiburg (48, 49). Subjects eligible for participation were German-speaking women residing in the two study areas and having in situ or invasive breast cancer diagnosed at less than age 51 years. Of the 1,005 living eligible cases identified by monitoring hospital admissions, surgery schedules, and pathology records from 38 surveyed hospitals that serve the population of the two regions, 706 (70.2 percent) took part. Of the patients who did not participate, 51 (5.1 percent) did not because physicians refused to allow contact, 11 (1.1 percent) had health problems, 152 (15.1 percent) refused to participate, and 85 (8.5 percent) did not respond. Controls were selected from random lists of residents of the two study areas supplied by the population registries. For every recruited patient, two controls matched by exact age and study region were contacted by letter. Of the 2,257 eligible controls, 1,381 (61.2 percent) participated, 218 (9.7 percent) did not respond, and 658 (29.1 percent) refused to take part. All study participants gave their informed consent. The study was reviewed by the ethics committee of the Univer-

TABLE 1. Continued

Characteristic	Cases		Controls		Controls only: proportion classified as physically inactive	
	No.	%	No.	%	No.	Row %
Change in body mass index between ages 20 and 30 years						
Unchanged (±2)	233	64.7	570	64.3†	85	14.9
Increased	74	20.6	207	23.4	26	12.6
Decreased	34	9.4	46	5.2	6	13.0
Data missing	19	5.3	63	7.1	6	9.5
Parous						
No	66	18.3	177	20.0	40	22.6†
Yes	294	81.7	709	80.0	83	11.7
	Cases (mean (SE‡))		Controls (mean (SE))		Controls only (means of inactive/active*)	
Parous women						
Age at first birth (years)	24.1 (0.26)		24.6 (0.18)		27.5/24.2§	
Full-term pregnancies (no.)	1.8 (0.04)		1.9 (0.03)§		1.9/2.0	
Total duration of breastfeeding (months)	5.1 (0.46)		6.4 (0.34)§		7.4/6.3	
Age at menarche (years)	13.1 (0.08)		13.1 (0.05)		13.1/13.2	
Body mass index at age 20 years (kg/m^2)	20.9 (0.14)		20.8 (0.09)		20.7/20.8	
Weight at age 20 years (kg)	57.1 (0.42)		56.3 (0.26)		56.4/56.3	
Height (cm)	165.2 (0.32)		164.6 (0.20)		165.1/164.5	
Daily average alcohol intake (g)	8.7 (0.77)		6.7 (0.28)§		7.9/6.5	
Energy intake (kcal/day) (subgroup: 211 cases, 536 controls)	2,495.1 (72.3)		2,419.7 (45.5)		2,092.6/2,476.2§	

* Women were classified as inactive if their total metabolic equivalent (MET)-hours/week values were in the first quartile during each age period (12–19 years, 20–30 years). All other women were classified as active.

† Significant difference between cases and controls or between active and inactive controls, respectively; $p < 0.05$ (chi-square test).

‡ SE, standard error.

§ Significant difference between cases and controls or between active and inactive controls, respectively; $p < 0.05$ (*t* test).

sity of Heidelberg. The participants completed a self-administered questionnaire assessing demographic, reproductive, menstrual, and anthropometric factors; family history of cancer; oral contraceptive use; medical history; diet; smoking; and alcohol consumption.

To obtain detailed information on physical activity, all participants were recontacted by letter in August 1999 and were invited to take part in a computer-assisted telephone interview. A follow-up conducted through the population registries to obtain changed addresses also yielded the date of death of the 115 deceased cases and three deceased controls. The interviews were conducted from September 1999 to May 2000 by trained interviewers blinded to the case-control status of the participant. Of the 572 cases and 1,353 controls for whom valid telephone numbers could be ascertained, 81.8 percent of the cases and 81.2 percent of the controls gave an interview, 10.3 percent of the cases and 12.9 percent of the controls refused to participate, and 7.9 percent of the cases and 6.0 percent of the controls could not be reached. In summary, 468 of the cases (46.6 percent of the original eligible cases or 66.3 percent of the original study population) and 1,093 of the controls (48.4 percent of the eligible controls or 79.1 percent of the original study control group) participated in the telephone investigation on physical activity.

Menopausal status was assigned according to the woman's reported status 6 months before the reference date. Women were defined as postmenopausal if they reported a natural menopause or a bilateral oophorectomy. The menopausal status of women who reported hysterectomy alone was classified as unknown. In total, 360 (76.9 percent) of the 468 cases and 886 (81.1 percent) of the 1,093 controls were premenopausal, 6.8 percent of the cases and 6.1 percent of the controls were postmenopausal, and, for 16.2 percent of

the cases and 12.9 percent of the controls, menopausal status was unknown. All analyses that follow were restricted to the subgroup of 1,246 clearly premenopausal women.

Assessment of physical activity

Physical activity was assessed during adolescence (age 12–19 years) and young adulthood (age 20–30 years) by using a detailed questionnaire on walking, cycling, household tasks, and 41 different sports activities. The frequency and duration of these activities were assessed by recording the numbers of years, months per year, days per week, and hours per day that each activity was performed. The intensity of the activity was estimated by the participant as light, moderate, heavy, or varying from instance to instance. In addition, lifetime occupational activity was assessed for all jobs held for more than 1 year. The years of starting and stopping the job, and whether it was full or part time, were reported. For each job period, intensity of the activity was assessed as mainly sedentary, standing/walking, or strenuous. Sedentary occupations were classified as physical inactivity. A specific metabolic equivalent (MET) abstracted from the Compendium of Physical Activities (50, 51) was assigned to each activity according to its reported intensity. A MET is defined as the ratio of the metabolic rate associated with a specific activity to the resting metabolic rate. The variables estimated for the analyses were expressed in MET-hours/week, calculated as the mean number of hours per week that participants engaged in an activity weighted by that activity's MET over both age periods separately and combined. Total physical activity was estimated as the sum of all reported activities.

Statistical analysis

Beyond standard descriptive statistics, we used logistic regression modeling conditioned on the matching variable age to estimate the odds ratios associated with breast cancer. We analyzed the different types of physical activities categorized via quartiles of the MET scores (MET-hours/week) of the controls separately in univariate models and in a multivariate model including all activities, as well as total activity. We investigated activity during adolescence and adulthood separately, in both periods together as well as in combination, categorized as inactive in both periods, active in both periods, active in adolescence only, and active in adulthood only. Linear trends were tested by entering physical activities as continuous variables in the models. To investigate interactions between different activities, we analyzed models that included a single activity (e.g., sports) and the sum of all other activities and the interaction term of both variables (not all data shown).

Potential confounding variables were investigated by using the difference between deviances of the models with and without the variable. The final models were adjusted for those variables found to influence goodness of fit: first-degree family history of breast cancer (yes/no), number of full-term pregnancies (0, 1, 2, ≥3), height (continuous), change in body mass index (kg/m^2) between ages 20 and 30 years (unchanged ±2, increase, decrease), total months of breastfeeding (0, 1–6, ≥7), and mean daily alcohol consumption (0, 1–18, ≥19 g). Other factors such as study region; level of education; body mass index at age 20, 30, and 40 years; body shape at menarche assessed as a choice of seven silhouettes; weight at age 20 years; age at menarche; age at first birth; or use of oral contraception did not influence the estimates for physical activity.

RESULTS

Table 1 summarizes some relevant characteristics and potential confounders for the study population of cases and controls. The mean age of cases was 41.9 years and that of controls was 42.5 years. When cases and controls were compared, controls were found to have a lower occurrence of family history of breast cancer, greater increase in body mass index between ages 20 and 30 years, more full-term pregnancies, longer durations of breastfeeding, and lower daily average alcohol consumption. Data on age at menarche, oral contraceptive use, age at first birth, educational level, body mass index and weight at age 20 years, height, energy intake, and smoking status were similarly distributed between cases and controls. Further classification of the population controls according to their total physical activity level during each age period (table 1) showed that women who reported low total physical activity were more likely to be nulliparous and have a higher educational level, a higher age at first birth, and a lower energy intake.

Physical activity is a multidimensional exposure variable. For each type of physical activity, table 2 describes the distribution of the summary measure for physical activity in MET-hours/week. At age 12–19 years, the highest contribution to total physical activity was from walking, followed by sports and cycling. Household activities were less relevant, and occupational activities had no impact at that age. At age 20–30 years, the highest activity levels stemmed from household and walking, followed by occupational activities. Combining both age intervals resulted in the main contributions resulting from walking and household, followed by occupational activities. During adolescence, 96 percent of all women participated in any sports, mainly at school (92 percent), followed by swimming (19 percent) and gymnastics (11 percent). At age 20–30 years, only 54 percent of all women reported any sports, mainly gymnastics (17 percent) and swimming (15 percent) (data not shown). In general, this crude comparison showed only minor differences between cases and controls.

Table 3 gives the odds ratios for breast cancer and level of total physical activity in MET-hours/week for the two time periods separately and combined. No significant effects of total physical activity were observed for the separate age periods. When activity patterns were studied over time and those women whose total activity levels were below the period-specific median in both age periods were used as the reference group, women who were physically active in adolescence had only slightly reduced risks, and women who continued their activities throughout life had the highest risk reductions (adjusted odds ratio (OR) = 0.83, 95 percent confidence interval (CI): 0.60, 1.14). However, none of these reductions was statistically significant. After both periods

TABLE 2. Distribution of physical activity in cases and controls at different ages based on MET*-hours/week for different types of physical activity assessed in a population-based case-control study of premenopausal breast cancer in Germany, 1999–2000

Type of physical activity	Median (25% quartile, 75% quartile) value in MET-hours/week					
	Adolescence (age 12–19 years)		Early adulthood (age 20–30 years)		Both periods (age 12–30 years)	
	Cases	Controls	Cases	Controls	Cases	Controls
Walking	23.8 (16.5, 45.6)	23.8 (14.3, 46.2)	21.6 (10.4, 34.7)	21.0 (9.9, 36.0)	25.0 (14.7, 36.5)	23.4 (13.9, 38.9)
Cycling	10.5 (1.0, 24.0)	11.8 (2.3, 29.2)	3.1 (0.0, 14.0)	4.1 (0.0, 15.9)	8.1 (2.0, 18.2)	10.3 (3.0, 21.0)
Sports	10.8 (5.2, 24.0)	12.0 (5.6, 25.0)	0.6 (0.0, 8.7)	1.9 (0.0, 11.2)	7.1 (3.2, 16.3)	7.9 (3.4, 18.8)
Household	7.5 (0.9, 17.5)	6.3 (0.0, 17.5)	35.0 (16.0, 113.8)	31.0 (15.0, 32.0)	24.4 (12.5, 45.8)	22.0 (11.2, 43.4)
Occupation†	0.0 (0.0, 26.7)	0.0 (0.0, 23.8)	6.5 (0.0, 45.1)	13.0 (0.0, 47.5)	12.5 (0.0, 34.4)	13.8 (0.0, 35.0)
Total	84.1 (55.0, 123.2)	88.7 (55.7, 134.0)	105.1 (64.5, 158.4)	109.0 (69.0, 160.4)	98.0 (67.2, 146.8)	104.0 (70.4, 145.5)

* MET, metabolic equivalent.

† Only information about the starting and stopping year of each occupation and whether the job was full or part time, mainly sedentary, standing/walking, or physically strenuous was assessed in the questionnaire. Sedentary jobs were not considered physical activity (MET = 0). A full-time job was assumed to be 38 hours/week, and part time was considered 19 hours/week.

were combined, all higher levels of total activity were associated with reduced odds ratios compared with low levels of physical activity (lowest quartile) in the multivariate adjusted model; however, only those for moderately high physical activity were statistically significant, that is, the third quartile of 104.1–145.5 MET-hours/week (adjusted OR = 0.68, 95 percent CI: 0.46, 0.99). We did not observe a trend in risk of breast cancer associated with increasing levels of total physical activity. The risk estimates were not substantially changed after adjustment for possible confounding factors.

Table 4 gives the odds ratios for breast cancer by physical activity type and level in MET-hours/week for the two time periods combined. For physical activity via cycling, we found decreasing risks with increasing cycling activity levels (trend-test $p = 0.03$). The risk for women who reported the highest levels of cycling activities was reduced in all models, with an adjusted risk estimate of 0.66 (95 percent CI: 0.45, 0.97). Moderately high walking activity was associated with an increased risk of 1.48 (95 percent CI: 1.02, 2.13) after adjustment. For occupational physical activity, household tasks, and sports, we did not find any association with breast cancer risk. Adjustment for possible confounding factors did not change the risk estimates substantially.

Many studies in the literature have investigated the relation between physical activity and breast cancer by looking at sports only. We further analyzed the association between sports and premenopausal breast cancer by taking into account all other physical activities and possible interactions (table 5). Significant interactions were found. Women who reported low (below the median) activity levels for sports and low (below the median) activity levels for all other physical activities combined were defined as the reference group. Participation in sports during both age periods resulted in significant risk reductions of comparable magnitude for women who were also physically active besides participating in sports (OR = 0.46, 95 percent CI: 0.23, 0.95) and for women who were physically inactive in addition to sports (OR = 0.44, 95 percent CI: 0.22, 0.90). An even more pronounced risk reduction was found for women who did not participate in any sports during both age periods but who had high levels of all other types of physical activities (OR = 0.22, 95 percent CI: 0.08, 0.56).

It is conceivable that certain dietary factors may act as confounders or effect modifiers. For subjects from one of the two areas of our study (211 cases and 536 controls), a detailed assessment of nutritional habits based on a food frequency questionnaire with 148 items referring to 1 year prior to diagnosis/interview was available. Cases and controls had comparable energy intakes. Subgroup analyses (not shown) did not reveal any confounding or modifying effects between physical activity and total energy intake, variation in body mass index, or weight changes. Our findings also were not substantially modified among overweight women or lean women.

DISCUSSION

In the present study, associations with premenopausal breast cancer risk were examined for different types of physical activity for ages 12–19 and 20–30 years. No significant effects for total physical activity were found in any one of the age periods. Combining both age periods, we found a significant risk reduction for total physical activity at a moderately high level (104.1–145.5 MET-hours/week), but no reduction was found for the highest quartile. For cycling, we found a significant protective effect, and sports appeared to have some protective effect among otherwise less active women. For the other types of physical activity, only weak or no associations with breast cancer risk were observed. Before discussing these results, we will address strengths and limitations of this study.

Measuring physical activity in epidemiologic studies is difficult, and the use of different methods may partly explain the inconsistent results across studies. Our detailed assessment of nonoccupational activities (housework), occupation, transportation (walking and cycling), and sports with respect to the frequency, duration, and intensity during two time periods of adolescence and young adulthood improves upon many previously used instruments. To our knowledge, only

TABLE 3. Univariate and multivariate odds ratios and 95% confidence intervals for breast cancer for total physical activity (MET*-hours/week) for both age periods separately and combined in a population-based case-control study of premenopausal breast cancer in Germany, 1999–2000

Characteristic	Cases (n = 360)	Controls (n = 886)	Unadjusted		Multivariate adjusted†	
			OR*	95% CI*	OR	95% CI
Age 12–19 years (MET-hours/week)						
13.0–55.7	92	221	1.00		1.00	
55.8–88.7	102	221	1.10	0.78, 1.55	1.07	0.75, 1.52
88.8–134.0	96	221	1.03	0.73, 1.45	1.00	0.70, 1.42
134.1–695.9	69	220	0.77	0.53, 1.11	0.73	0.50, 1.07
Data missing	1	3				
p for trend				0.44		0.44
Age 20–30 years (MET-hours/week)						
6.4–69.0	98	220	1.00		1.00	
69.1–109.0	90	220	0.93	0.66, 1.31	0.95	0.67, 1.37
109.1–160.4	83	220	0.87	0.61, 1.23	0.85	0.59, 1.23
160.5–728.8	88	220	0.89	0.63, 126	0.96	0.67, 1.39
Data missing	1	6				
p for trend				0.21		0.32
Both periods (age 12–30 years) (MET-hours/week)						
17.2–70.4	98	220	1.00		1.00	
70.5–104.0	102	220	1.05	0.74, 1.47	0.97	0.68, 1.38
104.1–145.5	68	220	0.70	0.49, 1.01	0.68	0.46, 0.99
145.6–564.4	91	220	0.92	0.65, 1.31	0.94	0.65, 1.35
Data missing	1	6				
p for trend				0.21		0.29
Total activity above the median as an adolescent/as an adult						
No/no	131	295	1.00		1.00	
Yes/no	57	145	0.90	0.62, 1.30	0.85	0.58, 1.25
No/yes	63	145	0.99	0.68, 1.43	0.99	0.68, 1.46
Yes/yes	108	295	0.83	0.61, 1.13	0.83	0.60, 1.14
Data missing	1	6				

* MET, metabolic equivalent; OR, odds ratio; CI, confidence interval.

† Model adjusted for first-degree family history of breast cancer (yes/no), number of full-term pregnancies (0, 1, 2, ≥3), height (continuous), change in body mass index between age 20 and 30 years (unchanged ±2, increase, decrease), total months of breastfeeding (0, 1–6, ≥7), and mean daily alcohol consumption (0, 1–18, ≥19 g).

two studies (30, 31) had comparable or more detailed coverage of physical activity with regard to all types of activities and assessed periods of participants' lifetime.

Selection bias cannot be excluded in this study since women were recontacted for a telephone interview and not all women participated in the physical activity evaluation. We had only limited information on nonresponders, but they might have had a less healthful lifestyle, possibly leading to an artificially high proportion of physically active women in our control group. However, for other lifestyle factors, such as smoking, alcohol consumption, and social variables, nonparticipants did not differ substantially from participating cases and controls.

We observed little or no confounding in our analyses, even after controlling for most known breast cancer risk factors. In our study, active women and inactive women had similar smoking and drinking habits. Subgroup analyses did not reveal modifying effects of smoking habits or alcohol consumption. These findings are consistent with those from almost all other studies. Only one study (30) reported modifying effects of smoking and drinking. Our results indicating no confounding or effect modification of anthropometric and certain dietary factors on the association between physical

TABLE 4. Univariate and multivariate odds ratios and 95% confidence intervals for breast cancer by physical activity type and level (MET*-hours/week) in both age periods (12–19 years, 20–30 years) combined in a population-based case-control study of premenopausal breast cancer in Germany, 1999–2000

Type of activity and quartiles of MET-hours/week (both periods)	Cases (no.)	Controls (no.)	Univariate		Multivariate unadjusted†		Multivariate adjusted‡	
			OR*	95% CI*	OR	95% CI	OR	95% CI
Walking								
0.0–13.9	86	221	1.00		1.00		1.00	
14.0–23.4	75	221	0.91	0.63, 1.31	0.93	0.64, 1.35	0.88	0.60, 1.29
23.5–38.9	121	221	1.43	1.02, 2.01	1.48	1.04, 2.10	1.48	1.02, 2.13
39.0–237.3	77	220	0.93	0.64, 1.33	0.94	0.64, 1.38	0.93	0.63, 1.39
p for trend				0.51		0.75		0.67
Cycling								
0.0–3.0	106	221	1.00		1.00		1.00	
3.1–10.3	96	223	0.89	0.64, 1.25	0.89	0.63, 1.26	0.89	0.62, 1.27
10.4–21.0	85	219	0.81	0.57, 1.14	0.81	0.57, 1.15	0.78	0.54, 1.13
21.1–163.9	72	220	0.67	0.47, 0.96	0.65	0.45, 0.95	0.66	0.45, 0.97
p for trend				0.02		0.02		0.03
Sports								
0.0–3.4	92	220	1.00		1.00		1.00	
3.5–7.9	101	220	1.10	0.78, 1.55	1.10	0.77, 1.56	1.17	0.81, 1.69
8.0–18.8	91	220	1.03	0.73, 1.46	1.08	0.75, 1.56	1.14	0.78, 1.67
18.9–143.7	75	220	0.83	0.58, 1.20	0.90	0.61, 1.32	0.89	0.60, 1.33
p for trend				0.85		0.66		0.92
Household								
0.0–11.2	80	223	1.00		1.00		1.00	
11.3–22.0	78	219	0.96	0.67, 1.39	0.95	0.66, 1.39	0.97	0.66, 1.42
22.1–43.4	100	221	1.25	0.88, 1.77	1.23	0.86, 1.77	1.28	0.87, 1.87
43.5–273.0	101	220	1.28	0.90, 1.82	1.30	0.89, 1.89	1.48	0.98, 2.23
p for trend				0.86		0.62		0.23
Occupation								
0.0	139	310	1.00		1.00		1.00	
>0.0–13.8	48	134	0.81	0.55, 1.20	0.79	0.53, 1.17	0.76	0.51, 1.15
13.9–35.0	95	231	0.90	0.66, 1.24	0.89	0.64, 1.23	0.91	0.65, 1.28
35.1–170.5	78	211	0.85	0.61, 1.18	0.86	0.61, 1.20	0.83	0.59, 1.18
p for trend				0.50		0.65		0.56

* MET, metabolic equivalent; OR, odds ratio; CI, confidence interval.

† Model includes all types of activity.

‡ Model was adjusted for all types of activity and additionally for first-degree family history of breast cancer (yes/no), number of full-term pregnancies (0, 1, 2, ≥3), height (continuous), change in body mass index between age 20 and 30 years (unchanged ±2, increase, decrease), total months of breastfeeding (0, 1–6, ≥7), and mean daily alcohol consumption (0, 1–18, ≥19 g).

activity and breast cancer risk are plausible given that premenopausal obesity or weight gain is not a risk factor for breast cancer (45).

We restricted this paper to premenopausal women only (79.8 percent of the total study population). Valid subgroup analyses for postmenopausal women were not possible. Comparable analyses for the total study population yielded similar results (data not shown). In the literature, findings on premenopausal women are conflicting. Of all 20 studies that reported on premenopausal women, including three cohort (5, 16, 47) and 17 case-control (18, 20, 22, 25, 26, 29–34, 37–40, 42, 43) studies, six (5, 18, 20, 26, 31, 38) observed a significant inverse association between some type of physical activity and breast cancer risk, six (22, 25, 29, 33, 39, 40) reported supportive but nonsignificant results for an inverse association, and eight (16, 30, 32, 34, 37, 42, 43, 47)

TABLE 5. Adjusted odds ratios* and 95% confidence intervals for breast cancer by sports activity pattern over both age periods (12–19 years, 20–30 years) stratified by the degree of physical activity in all other activities besides sports, population-based case-control study of premenopausal breast cancer in Germany, 1999–2000

Active in sports as adolescent/adult	Degree of activity in all other physical activities besides sports†							
	Inactive (≤median)				Active (>median)			
	Cases	Controls	OR‡	95% CI‡	Cases	Controls	OR	95% CI
No/no	17	22	1.00		11	51	0.22	0.08, 0.56
Yes/no	71	158	0.52	0.25, 1.08	73	158	0.60	0.29, 1.23
No/yes	7	17	0.54	0.18, 1.66	7	18	0.36	0.12, 1.12
Yes/yes	97	243	0.46	0.23, 0.95	76	213	0.44	0.22, 0.90

* Model was adjusted for first-degree family history of breast cancer (yes/no), number of full-term pregnancies (0, 1, 2, ≥3), height (continuous), change in body mass index between age 20 and 30 years (unchanged ±2, increase, decrease), total months of breastfeeding (0, 1–6, ≥7), and mean daily alcohol consumption (0, 1–18, ≥19 g).

† Mean metabolic-equivalent score over both age periods for all physical activities except sports. Women were classified as inactive if their activity was below the median value and as active otherwise.

‡ OR, odds ratio; CI, confidence interval.

did not find any association. The six studies that found an association yielded risk reductions in the range of 0.42 (18, 26) to 0.74 (20). Of the two largest studies (33, 34) with more than 1,000 premenopausal breast cancer cases, one (33) was supportive and the other (34) did not find an association. Of the two studies with the most detailed exposure assessment (30, 31), one (31) observed a significant inverse association and the other did not find any association between premenopausal breast cancer risk and physical activity.

We collected information on physical activity for two different time periods. We did not find any different effects of total physical activity by time period. With regard to activity patterns over time, women who were physically active during adolescence had slightly reduced risks, and women who continued their activities throughout life had the highest risk reductions. However, none of these reductions was statistically significant. We know of only four other publications (29, 31, 42, 52) that have examined the independent effects of physical activity in adolescence and in adulthood as well as changes in activity over time. Three of them (29, 31, 52) also found most pronounced risk reductions for women who exercised throughout life. However, in contrast to our findings, these studies reported a larger risk reduction for women who initiated exercise in adulthood compared with those who were active in adolescence only. One study did not find any effects of different time periods (42).

To our knowledge, this is the first case-control study to investigate physical activity and breast cancer risk in Germany. Because physical activity may be associated with different national habits, it may be informative to perform studies in different countries. For example, in Germany, cycling is not only a recreational activity but also widely used as a means of transportation. As such, our study population differs from others, for instance, from North America. In our study, walking and cycling accounted for one third of total activity, and cycling as a means of transportation made a higher contribution than sports to the total activity score. For cycling, we observed a clear duration-response and dose-response relation with regard to MET-hours/week. For women in the highest quartile (>21 MET-hours/week, i.e., women who cycled more than 3 hours/week with moderate intensity, for example), a protective effect of 34 percent was found. Very few studies have published data on cycling, and sometimes walking and cycling have been reported together. In a study performed in the Netherlands on postmenopausal women, a relative risk of 0.81 (95 percent CI: 0.60, 1.09) was reported for women who cycled or walked daily for more than 1 hour compared with women who did so for less than 10 minutes a day (9). In a Finnish study, women who reported commuting, walking, or bicycling to work 30 minutes or more daily had a slightly lower risk of breast cancer (relative risk = 0.87, 95 percent CI: 0.62, 1.24) than women working at home, being unemployed, or driving a car to work (17). Our findings support the suggestion by Ainsworth et al. (46) that nonoccupational activity, such as commuting to work, may be important for breast cancer risk, depending on the culture of physical activity in a country. A possible explanation for why cycling demonstrated a clear trend when other activities did not is that cycling is a salient activity that may be recalled better than other activities performed less frequently. Hence, there may have been less measurement error for this variable.

For sports in adolescence and/or adulthood, we did not find an effect on breast cancer risk in the univariate and multivariate models. However, in the more complex models, we found that sports for otherwise less active women resulted in a breast cancer risk reduction of about 50 percent. Confirmation of this result would be important for public health recommendations. For otherwise active women, participating in additional sports does not induce a further risk reduction. Some of the studies that assessed sports activity observed a reduction in breast cancer risk (1, 5, 6, 9, 18, 21, 23, 29, 31, 38, 40, 42), whereas others found no association (14, 30, 47). One case-control study found an increased breast cancer risk (36).

We did not find a clear trend or overall monotonic effect of decreasing breast cancer risk with increasing total physical activity that detracts from a causal interpretation. However, our findings on moderately high levels of total activity are consistent with those from some other studies that found protective effects for moderate activity but no effects or less-pronounced effects for high activity levels (9, 29, 42, 53). It is possible that the association is not linear but may have a threshold or a U-shape (54). These findings might be biologically plausible because moderate physical activity may enhance the immune system by elevating the number of natural killer cells, whereas exhaustive physical exercise may depress immunologic functions (55, 56). Furthermore, activities performed at a vigorous level of intensity may be associated with hormone production and uptake, and/or prostaglandin production and uptake, and disturbances in overall energy balance (46, 55).

In conclusion, these data do not suggest an inverse monotonic association between total physical activity and breast cancer risk in premenopausal women. In our study, a moderately high physical activity level was associated with a decreased breast cancer risk in premenopausal women. Confirmation of this association is of great public health importance. Our findings of a protective effect of cycling emphasize that the assessment methods need to be comprehensive and adapted to national habits. In our study, one third of the total exposure would have been missed if walking and cycling for transportation reasons had not been included in the interview. Furthermore, our study revealed interesting findings on sports in interaction with other physical activities, demonstrating that not only more refined assessment methods but also more refined analyses might be useful to study different aspects of physical activity and their relation to breast cancer risk.

ACKNOWLEDGMENTS

This work was supported by the charity organization German Cancer Aid (Deutsche Krebshilfe e.V. Project 70-2225).

The authors thank the many gynecologists and oncologists in the 38 clinics of the study regions Rhein-Neckar-Odenwald and the greater area of Freiburg for allowing them to contact their patients; Ursula Eilber for competent data coordination and management; Dr. Silke Schieber, Dr. Andrea Busche-Bässler, Dr. Regina Hübner, Ruth Schäuble, Dr. Heike Wiedensohler, Renate Birr, Ulla Gromer, and Dr. Ulrike Bussas for data collection for the first questionnaire; and the Center for Survey Research and Methodology (ZUMA), Mannheim, for efficient interviews regarding the physical activity exposures.

This paper is dedicated to Professor Harald zur Hausen on the occasion of his retirement as head of the German Cancer Research Center (Deutsches Krebsforschungszentrum) in Heidelberg, with gratitude and appreciation for 20 years of leadership.

REFERENCES

1. Frisch RE, Wyshak G, Albright NL, et al. Lower prevalence of non-reproductive system cancers among female former college athletes. Med Sci Sports Exerc 1989;21:250–3.
2. Vena JE, Graham S, Zielezny M, et al. Occupational exercise and risk of cancer. Am J Clin Nutr 1987;45:318–27.
3. Zheng W, Shu XO, McLaughlin JK, et al. Occupational physical activity and the incidence of cancer of the breast, corpus uteri, and ovary in Shanghai. Cancer 1993;71:3620–4.
4. Fraser GE, Shavlik D. Risk factors, lifetime risk, and age at onset of breast cancer. Ann Epidemiol 1997;7:375–82.
5. Thune I, Brenn T, Lund E, et al. Physical activity and the risk of breast cancer. (See comments). N Engl J Med 1997;336:1269–75.
6. Rockhill B, Willett WC, Hunter DJ, et al. A prospective study of recreational physical activity and breast cancer risk. Arch Intern Med 1999;159:2290–6.
7. Moore DB, Folsom AR, Mink PJ, et al. Physical activity and incidence of postmenopausal breast cancer. Epidemiology 2000;11:292–6.
8. Wyshak G, Frisch RE. Breast cancer among former college athletes compared to non-athletes: a 15-year follow-up. Br J Cancer 2000;82:726–30.
9. Dirx MJ, Voorrips LE, Goldbohm RA, et al. Baseline recreational physical activity, history of sports participation, and postmenopausal breast carcinoma risk in the Netherlands Cohort Study. Cancer 2001;92:1638–49.
10. Pukkala E, Poskiparta M, Apter D, et al. Life-long physical activity and cancer risk among Finnish female teachers. Eur J Cancer Prev 1993;2:369–76.
11. Dorgan JF, Brown C, Barrett M, et al. Physical activity and risk of breast cancer in the Framingham Heart Study. Am J Epidemiol 1994;139:662–9.
12. Steenland K, Nowlin S, Palu S. Cancer incidence in the National Health and Nutrition Survey I. Follow-up data: diabetes, cholesterol, pulse and physical activity. Cancer Epidemiol Biomarkers Prev 1995;4:807–11.
13. Sesso HD, Paffenbarger RS Jr, Lee IM. Physical activity and breast cancer risk in the College Alumni Health Study (United States). Cancer Causes Control 1998;9:433–9.
14. Rockhill B, Willett WC, Hunter DJ, et al. Physical activity and breast cancer risk in a cohort of young women. (See comments). J Natl Cancer Inst 1998;90:1155–60.
15. Calle EE, Murphy TK, Rodriguez C, et al. Occupation and breast cancer mortality in a prospective cohort of US women. Am J Epidemiol 1998;148:191–7.
16. Lee IM, Rexrode KM, Cook NR, et al. Physical activity and breast cancer risk: the Women's Health Study (United States). Cancer Causes Control 2001;12:137–45.
17. Luoto R, Latikka P, Pukkala E, et al. The effect of physical activity on breast cancer risk: a cohort study of 30,548 women. Eur J Epidemiol 2000;16:973–80.
18. Bernstein L, Henderson BE, Hanisch R, et al. Physical exercise and reduced risk of breast cancer in young women. (See comments). J Natl Cancer Inst 1994;86:1403–8.
19. Mittendorf R, Longnecker MP, Newcomb PA, et al. Strenuous physical activity in young adulthood and risk of breast cancer (United States). Cancer Causes Control 1995;6:347–53.
20. Hirose K, Tajima K, Hamajima N, et al. A large-scale, hospital-based case-control study of risk factors of breast cancer according to menopausal status. Jpn J Cancer Res 1995;86:146–54.
21. McTiernan A, Stanford JL, Weiss NS, et al. Occurrence of breast cancer in relation to recreational exercise in women age 50–64 years. Epidemiology 1996;7:598–604.
22. Mezzetti M, La Vecchia C, Decarli A, et al. Population attribut-

able risk for breast cancer: diet, nutrition, and physical exercise. J Natl Cancer Inst 1998;90:389–94.
23. Marcus PM, Newman B, Moorman PG, et al. Physical activity at age 12 and adult breast cancer risk (United States). Cancer Causes Control 1999;10:293–302.
24. Carpenter CL, Ross RK, Paganini-Hill A, et al. Lifetime exercise activity and breast cancer risk among post-menopausal women. Br J Cancer 1999;80:1852–8.
25. Ueji M, Ueno E, Osei Hyiaman D, et al. Physical activity and the risk of breast cancer: a case-control study of Japanese women. J Epidemiol 1998;8:116–22.
26. Levi F, Pasche C, Lucchini F, et al. Occupational and leisure time physical activity and the risk of breast cancer. Eur J Cancer 1999;35:775–8.
27. Moradi T, Nyren O, Zack M, et al. Breast cancer risk and lifetime leisure-time and occupational physical activity (Sweden). Cancer Causes Control 2000;11:523–31.
28. Shoff SM, Newcomb PA, Trentham Dietz A, et al. Early-life physical activity and postmenopausal breast cancer: effect of body size and weight change. Cancer Epidemiol Biomarkers Prev 2000;9:591–5.
29. Verloop J, Rookus MA, van der Kooy K, et al. Physical activity and breast cancer risk in women aged 20–54 years. J Natl Cancer Inst 2000;92:128–35.
30. Friedenreich CM, Bryant HE, Courneya KS. Case-control study of lifetime physical activity and breast cancer risk. Am J Epidemiol 2001;154:336–47.
31. Matthews CE, Shu XO, Jin F, et al. Lifetime physical activity and breast cancer risk in the Shanghai Breast Cancer Study. Br J Cancer 2001;84:994–1001.
32. Gilliland FD, Li YF, Baumgartner K, et al. Physical activity and breast cancer risk in Hispanic and non-Hispanic White women. Am J Epidemiol 2001;154:442–50.
33. Coogan PF, Newcomb PA, Clapp RW, et al. Physical activity in usual occupation and risk of breast cancer (United States). Cancer Causes Control 1997;8:626–31.
34. Gammon MD, Schoenberg JB, Britton JA, et al. Recreational physical activity and breast cancer risk among women under age 45 years. Am J Epidemiol 1998;147:273–80.
35. Coogan PF, Aschengrau A. Occupational physical activity and breast cancer risk in the upper Cape Cod cancer incidence study. Am J Ind Med 1999;36:279–85.
36. Dosemeci M, Hayes RB, Vetter R, et al. Occupational physical activity, socioeconomic status, and risks of 15 cancer sites in Turkey. Cancer Causes Control 1993;4:313–21.
37. Taioli E, Barone J, Wynder EL. A case-control study on breast cancer and body mass. Eur J Cancer 1995;31A:723–8.
38. D'Avanzo B, Nanni O, La Vecchia C, et al. Physical activity and breast cancer risk. Cancer Epidemiol Biomarkers Prev 1996;5:155–60.
39. Hu YH, Nagata C, Shimizu H, et al. Association of body mass index, physical activity, and reproductive histories with breast cancer: a case-control study in Gifu, Japan. Breast Cancer Res Treat 1997;43:65–72.
40. Friedenreich CM, Rohan TE. Physical activity and risk of breast cancer. Eur J Cancer Prev 1995;4:145–51.
41. Coogan PF, Clapp RW, Newcomb PA, et al. Variation in female breast cancer risk by occupation. Am J Ind Med 1996; 30:430–7.
42. Lee IM, Cook NR, Rexrode KM, et al. Lifetime physical activity and risk of breast cancer. Br J Cancer 2001;85:962–5.
43. Chen CL, White E, Malone KE, et al. Leisure-time physical activity in relation to breast cancer among young women (Washington, United States). Cancer Causes Control 1997;8: 77–84.
44. Friedenreich CM. Physical activity and cancer prevention: from observational to intervention research. Cancer Epidemiol Biomarkers Prev 2001;10:287–301.
45. Weight control and physical activity. IARC handbooks of cancer prevention. Vol 6. Lyon, France: International Agency for Research on Cancer, 2002.
46. Ainsworth BE, Sternfeld B, Slattery ML, et al. Physical activity and breast cancer: evaluation of physical activity assessment methods. Cancer 1998;83:611–20.
47. Albanes D, Blair A, Taylor PR. Physical activity and risk of cancer in the NHANES I population. Am J Public Health 1989; 79:744–50.
48. Chang-Claude J, Eby N, Kiechle M, et al. Breastfeeding and breast cancer risk by age 50 among women in Germany. Cancer Causes Control 2000;11:687–95.
49. Kropp S, Becher H, Nieters A, et al. Low-to-moderate alcohol consumption and breast cancer risk by age 50 years among women in Germany. Am J Epidemiol 2001;154:624–34.
50. Ainsworth BE, Haskell WL, Leon AS, et al. Compendium of physical activities: classification of energy costs of human physical activities. Med Sci Sports Exerc 1993;25:71–80.
51. Ainsworth BE, Haskell WL, Whitt MC, et al. Compendium of physical activities: an update of activity codes and MET intensities. Med Sci Sports Exerc 2000;32:S498–504.
52. Friedenreich CM, Courneya KS, Bryant HE. Influence of physical activity in different age and life periods on the risk of breast cancer. Epidemiology 2001;12:604–12.
53. Friedenreich CM, Courneya KS, Bryant HE. Relation between intensity of physical activity and breast cancer risk reduction. Med Sci Sports Exerc 2001;33:1538–45.
54. Gammon MD, John EM, Britton JA. Recreational and occupational physical activities and risk of breast cancer. J Natl Cancer Inst 1998;90:100–17.
55. Hoffman Goetz L, Apter D, Demark Wahnefried W, et al. Possible mechanisms mediating an association between physical activity and breast cancer. Cancer 1998;83:621–8.
56. Bernstein L, Ross RK, Lobo RA, et al. The effects of moderate physical activity on menstrual cycle patterns in adolescence: implications for breast cancer prevention. Br J Cancer 1987;55: 681–5.